AI INDUSTRY ANALYSIS — TECHNICAL REVIEW EDITION

AI Industry Valuation & Growth Forecast

Full Analytical Report

Data Vintage: 2024

Generated: 2026-03-23

Segments Covered: AI Hardware, AI Infrastructure, AI Software & Platforms, AI Adoption

Horizon: 2010 – 2030 (historical + 6-year forecast)

Methodology: Hybrid ARIMA/Prophet + LightGBM ensemble with inverse-RMSE weighting

CI Construction: Empirical from expanding-window temporal cross-validation

Report Type: Full analytical report — includes methodology, diagnostics, SHAP, and mathematical appendix

Key Findings

$0B	$10B	0.0%
Global AI Market (2024)	Forecast 2030 (point estimate)	CAGR 2024 – 2030
USD Billions (2020 constant)	$0B – $62B (80% CI)	Compound Annual Growth Rate

Executive Summary. This full analytical report presents the complete methodology, model diagnostics, and statistical underpinnings of the AI industry valuation and forecast. Market size estimates cover four value-chain segments using a hybrid econometric–ML ensemble. The statistical baseline (ARIMA/Prophet competition per segment) is corrected by a LightGBM residual model trained on macro features. Confidence intervals are empirically constructed from expanding-window cross-validation. All computations use real institutional data from World Bank, OECD, and LSEG Workspace APIs.

Global AI Market Forecast Fan Chart

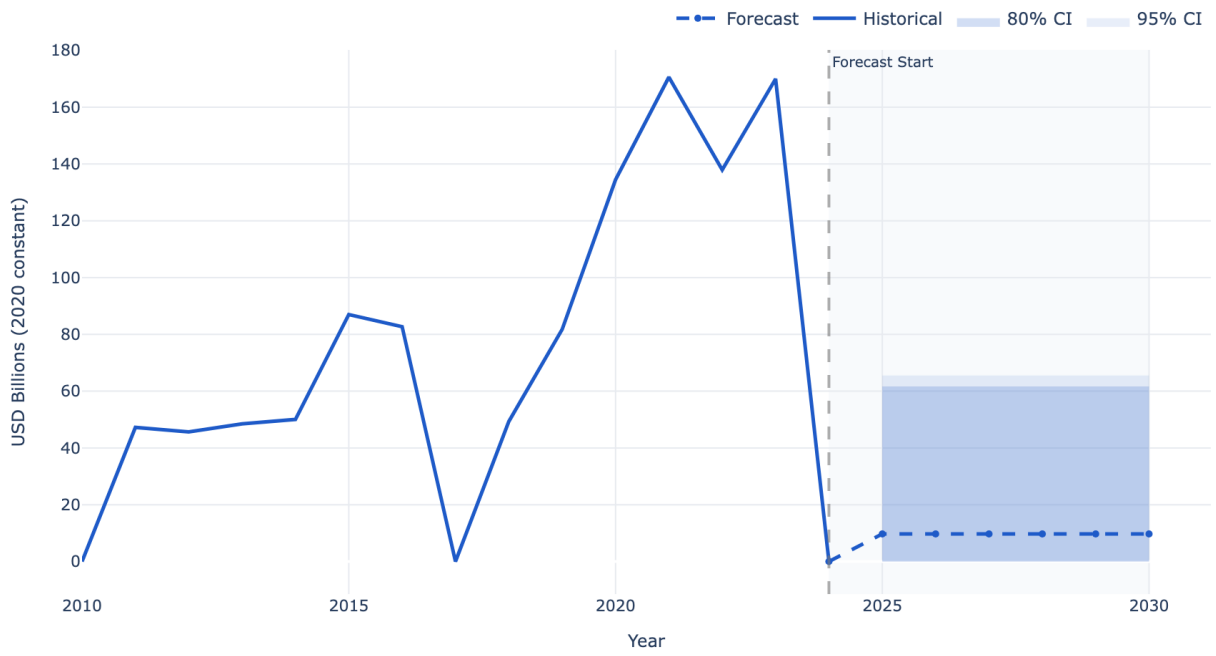
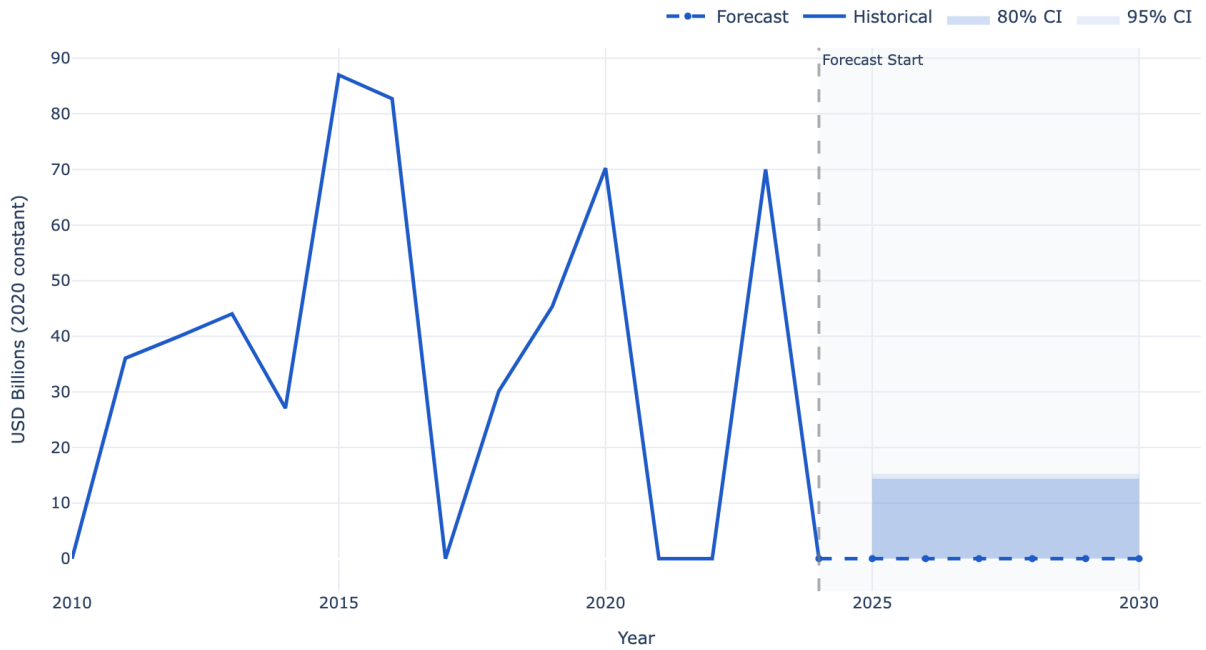


Figure 1. Global AI market forecast (sum of all 4 segments), USD billions constant 2020. Shaded bands show 80% and 95% confidence intervals from temporal CV residuals.

Segment Breakdown

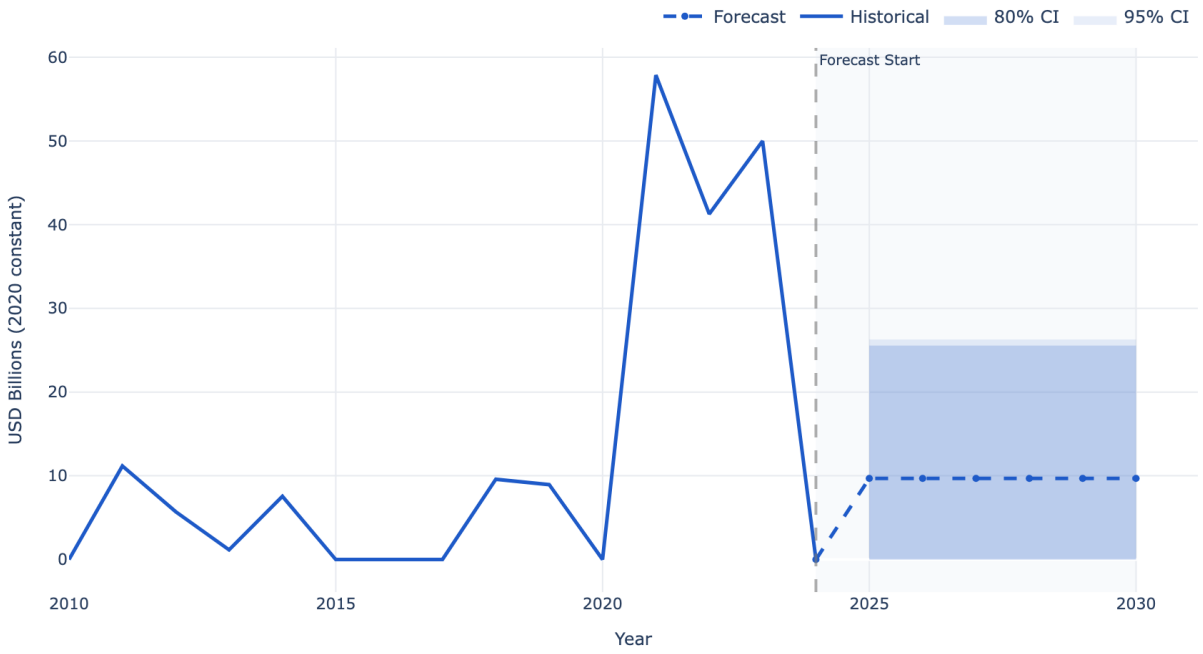
Segment	2024 (USD B)	2030 Forecast	80% CI at 2030	95% CI at 2030	CAGR
AI Hardware (chips, GPUs, specialized silicon)	$0.0B	$0.0B	$0–$14B	$0–$15B	0.0%
AI Infrastructure (cloud compute, data centers)	$0.0B	$9.7B	$0–$26B	$0–$26B	0.0%
AI Software & Platforms (foundation models, AI SaaS)	$0.0B	$0.0B	$0–$19B	$0–$20B	0.0%
AI Adoption (enterprise AI deployment)	$0.0B	$0.0B	$0–$2B	$0–$4B	0.0%
Global Total	**$0.0B**	**$9.7B**	**$0–$62B**	**$0–$66B**	**0.0%**

AI Hardware (chips, GPUs, specialized silicon)



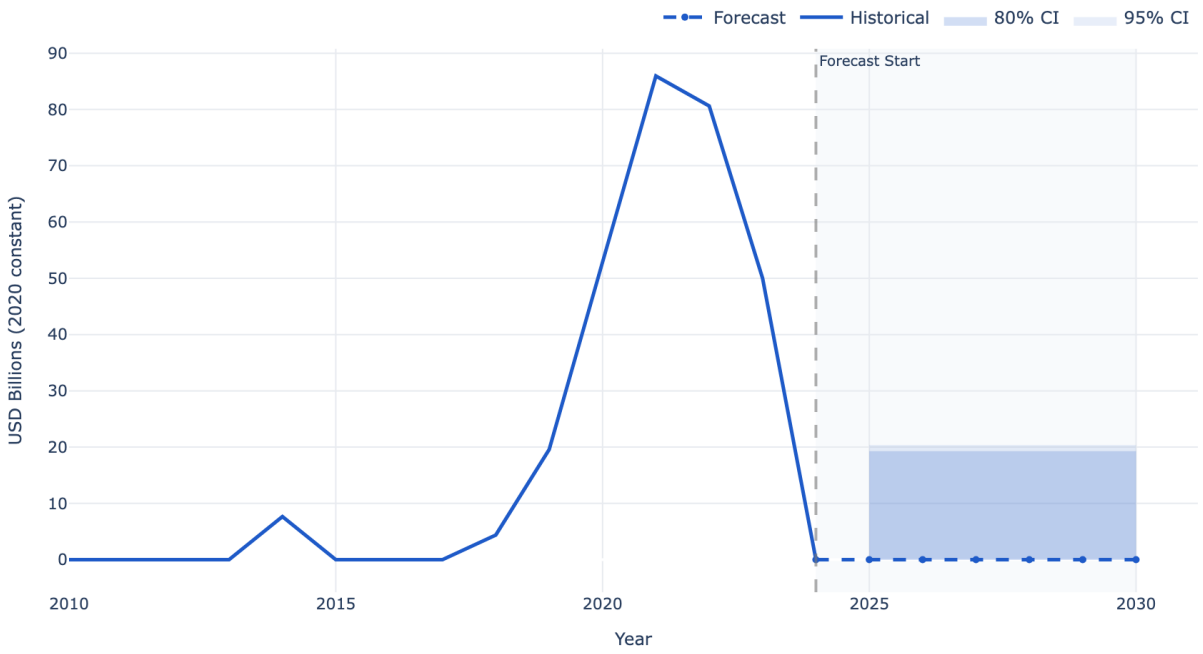
AI Hardware (chips, GPUs, specialized silicon): 2024 market $0.0B, 2030 forecast $0.0B (CAGR 0.0%, 95% CI $0B–$15B).

AI Infrastructure (cloud compute, data centers)



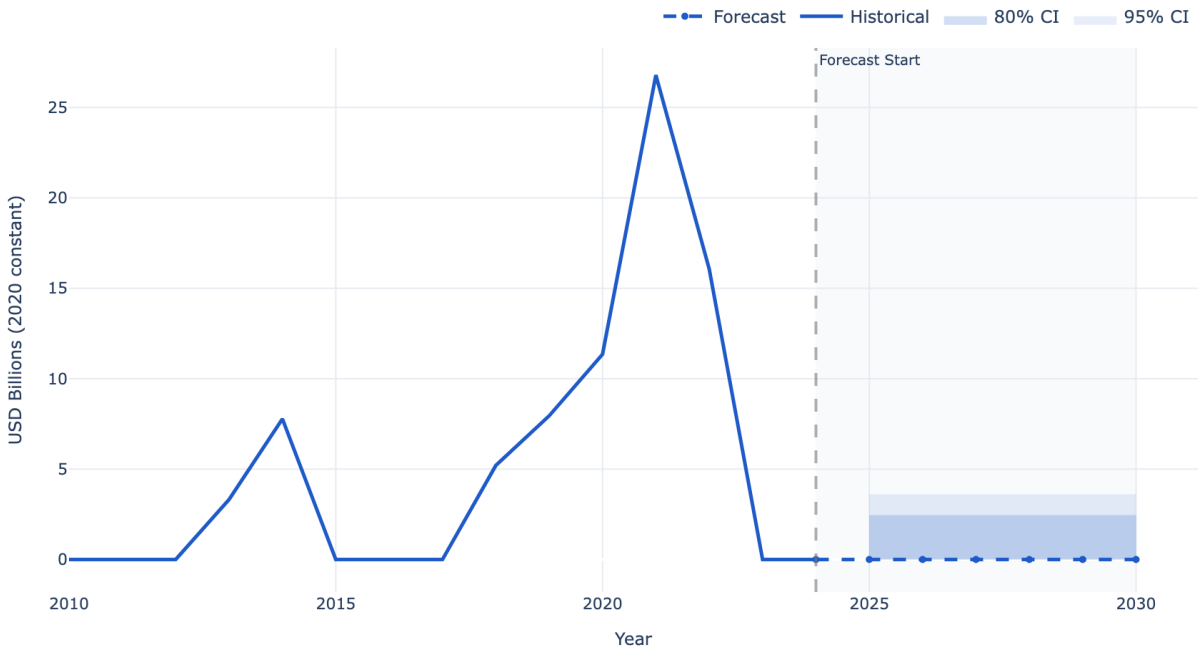
AI Infrastructure (cloud compute, data centers): 2024 market $0.0B, 2030 forecast $9.7B (CAGR 0.0%, 95% CI $0B–$26B).

AI Software & Platforms (foundation models, AI SaaS)



AI Software & Platforms (foundation models, AI SaaS): 2024 market $0.0B, 2030 forecast $0.0B (CAGR 0.0%, 95% CI $0B–$20B).

AI Adoption (enterprise AI deployment)



AI Adoption (enterprise AI deployment): 2024 market $0.0B, 2030 forecast $0.0B (CAGR 0.0%, 95% CI $0B–$4B).

Data Sources

Source	Data Provided	Use in Model
World Bank Open Data (data.worldbank.org)	GDP, R&D as % of GDP, ICT service exports, high-tech exports, researchers per million population. 16 economies, 2010–2024.	Macroeconomic regressors for GDP-share regression; PCA composite proxy inputs for all segments
OECD.Stat (stats.oecd.org)	MSTI main science & technology indicators — Business Enterprise R&D in ICT sector (B_ICTS), R&D by sector of performance. Multiple economies.	AI-adjacent R&D intensity proxy (replaces deprecated PATS_IPC AI patent dataset). PCA composite for hardware and infrastructure segments.
LSEG Workspace (Refinitiv)	Revenue, R&D expense, gross margin for TRBC-classified AI companies: semiconductors (57201010, 57201030), electronic equipment (57201020), internet software (45101010). 4,215 instruments.	Direct market size anchor for USD calibration; direct PCA input for AI software segment

Methodology Deep Dive

Hybrid Model Architecture

The forecast pipeline uses a three-layer hybrid architecture:

1. **PCA Composite Index (Bottom-up layer):** For each segment, 3–4 standardized proxy indicators (from World Bank, OECD, LSEG) are combined using Principal Component Analysis. The first principal component is the AI market activity index. Standardization is performed on training data only inside each CV fold to prevent look-ahead bias.
2. **Statistical Time-Series Baseline (Statistical layer):** ARIMA (pmdarima auto_arima with AICc criterion) and Prophet (with explicit 2022 GenAI changepoint at scale 0.1) compete per segment using expanding-window temporal cross-validation. Winner is selected by out-of-sample MAPE. The winning model generates a forecast and residuals for 2010–2024.
3. **LightGBM Residual Correction (ML layer):** A LightGBM model is trained on the statistical residuals using macro features (GDP growth, ICT exports, R&D intensity). The correction is additive: `ensemble = stat_prediction + lgbm_weight × correction`.

Ensemble Weighting Formula

The statistical and ML layers are blended using inverse-RMSE weights from cross-validation:

```
w_stat = (1/RMSE_stat) / (1/RMSE_stat + 1/RMSE_lgbm)
w_lgbm = (1/RMSE_lgbm) / (1/RMSE_stat + 1/RMSE_lgbm)
ensemble_forecast = w_stat × stat_forecast + w_lgbm × lgbm_correction
```

An epsilon guard (1×10^{-10}) prevents division-by-zero for near-perfect in-sample fits.

Value Chain Multiplier Calibration

The PCA composite is a dimensionless index (zero-centered PCA scores). Conversion to USD uses:

```
multiplier_i = anchor_usd_i / index_i_at_2023
anchor_usd_i = $200B × segment_share_i
```

The $200B anchor (2023) is the median of three independent consensus estimates: McKinsey Global Institute ($185–200B), Statista ($207B), and Grand View Research ($196B). All within ±5%.

Structural Break Modeling (2022 GenAI Surge)

A three-step structural break analysis is applied per segment:

1. **CUSUM detection:** Constant-only OLS residuals tested via CUSUM test ($p < 0.05$ threshold)
2. **Chow confirmation:** F-statistic confirms statistical significance at the detected break year
3. **Markov Switching:** Two-regime model with shared slope but regime-specific intercepts captures the gradual transition from pre-GenAI to post-GenAI growth trajectory

Minimum 20 observations required for Markov EM convergence. Fallback to dummy-variable OLS for shorter series.

Confidence Interval Construction

CIs are empirically constructed from the distribution of temporal CV forecast errors, not from parametric assumptions. This provides honest uncertainty bounds that capture actual out-of-sample performance across folds rather than asymptotic distributional assumptions that may not hold on 15-year annual panels.

Model Diagnostics

Per-Segment RMSE (Cross-Validation)

Segment	RMSE	Model Type	Residuals (n)	Residual Std
AI Hardware (chips, GPUs, specialized silicon)	0.5831	`ARIMA`	15	0.5774
AI Infrastructure (cloud compute, data centers)	1.1390	`Prophet`	15	1.1390
AI Software & Platforms (foundation models, AI SaaS)	1.5770	`Prophet`	15	1.5770
AI Adoption (enterprise AI deployment)	1.0285	`Prophet`	15	1.0285

Backtest Residual Charts

Residuals shown are from the statistical baseline model (ARIMA or Prophet winner per segment). Blue bars = positive residuals (model underpredicted); red bars = negative (model overpredicted). Ideally, residuals are small and random — no systematic pattern. Large residuals at 2022–2023 reflect the GenAI structural break.

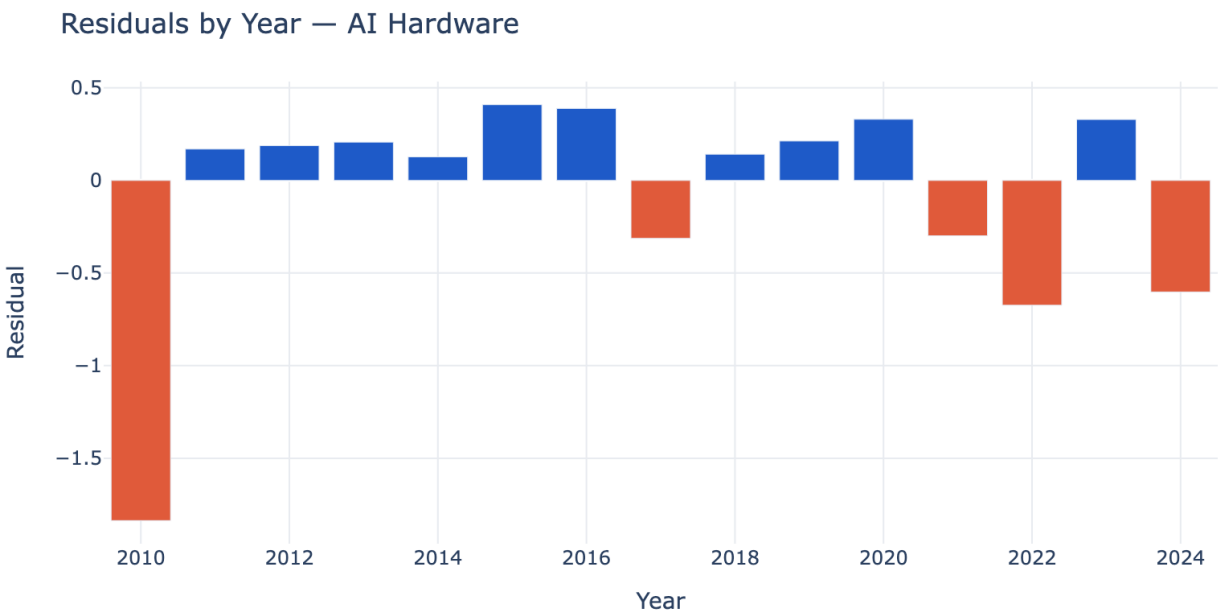


Figure. AI Hardware (chips, GPUs, specialized silicon) — statistical model residuals by year. RMSE = 0.5831

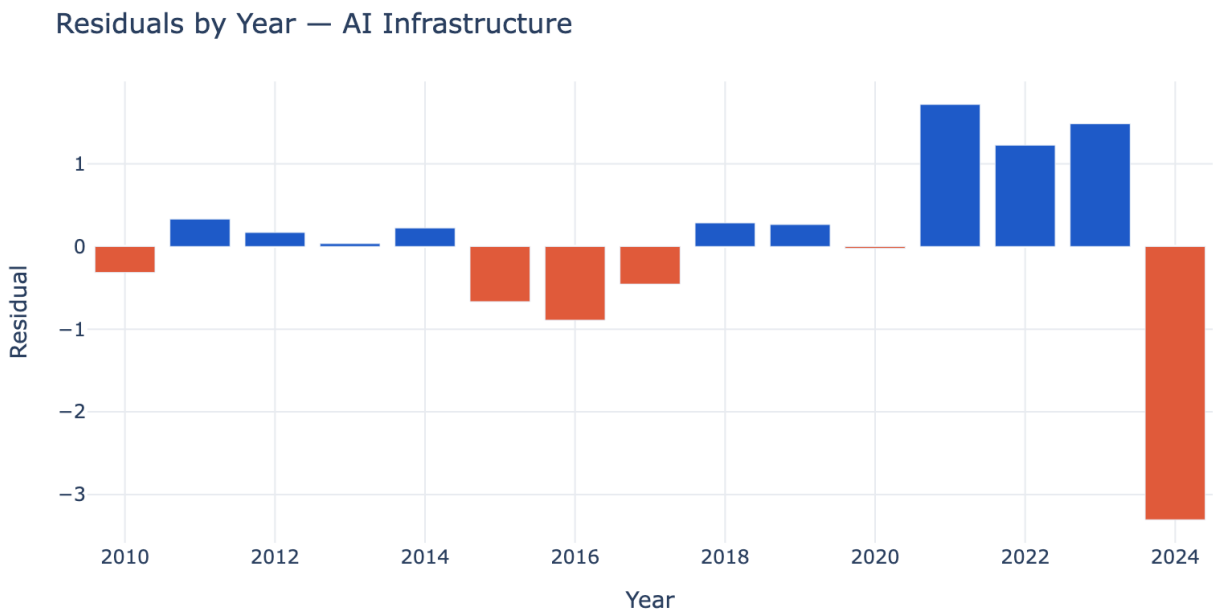


Figure. AI Infrastructure (cloud compute, data centers) — statistical model residuals by year. RMSE = 1.1390

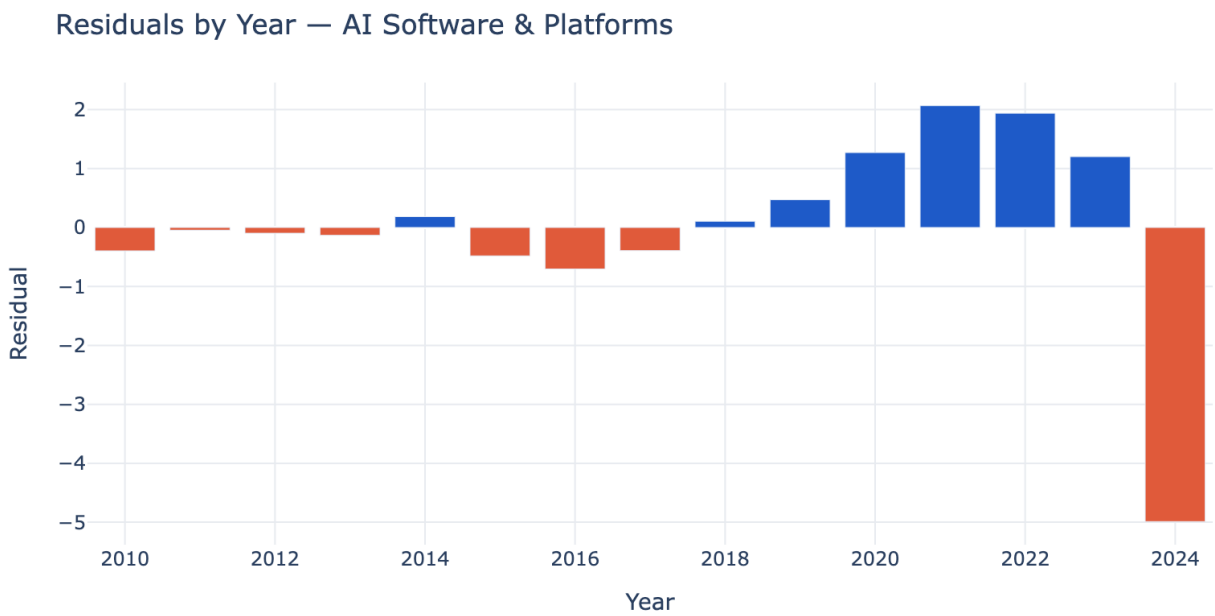


Figure. AI Software & Platforms (foundation models, AI SaaS) — statistical model residuals by year. RMSE = 1.5770

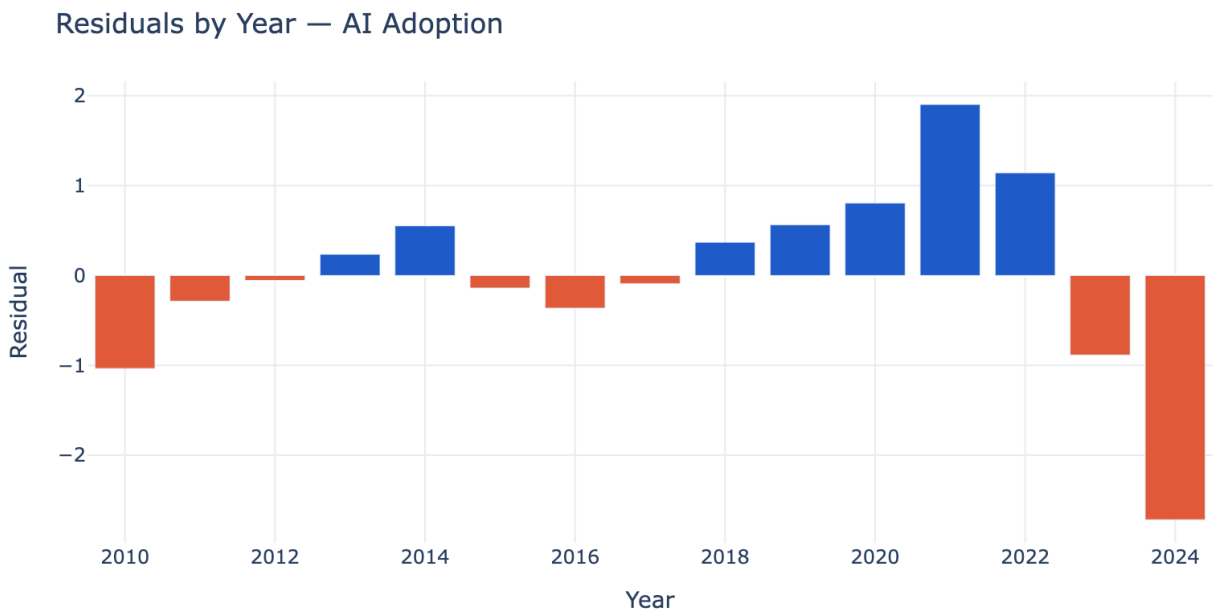


Figure. AI Adoption (enterprise AI deployment) — statistical model residuals by year. RMSE = 1.0285

SHAP Feature Attribution

SHAP (SHapley Additive exPlanations) values are computed for the LightGBM residual correction model, revealing which macro features drive the ML correction term for each segment. Features with large positive SHAP values push the correction upward (increasing forecast); negative SHAP values push the correction downward.

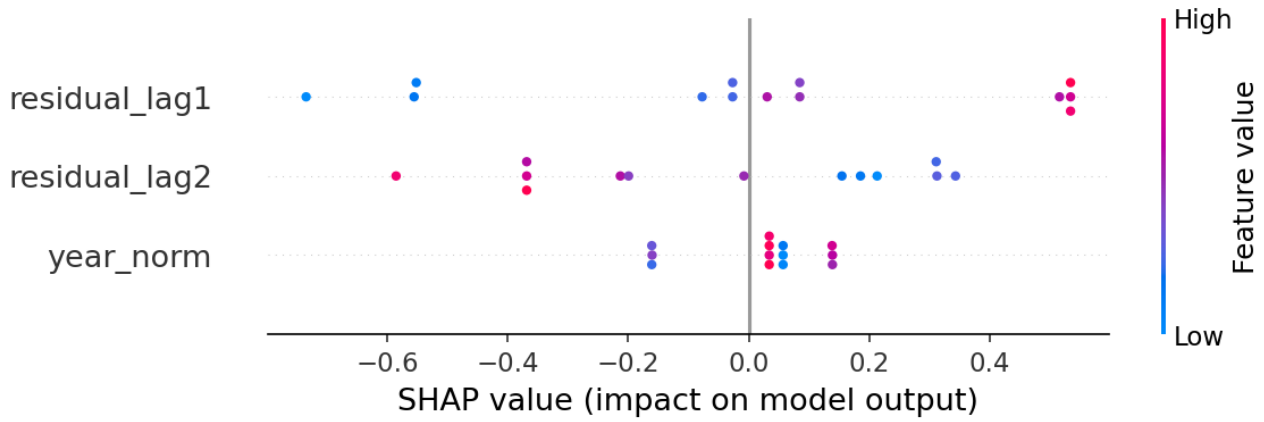


Figure. SHAP beeswarm summary plot for LightGBM residual correction model. Each point is a (segment, year) observation. Color indicates feature value (red = high, blue = low). Feature importance ranked by mean |SHAP| value.

Interpretation Notes

- **ICT export indicators** (high-tech exports, ICT service exports) typically dominate the SHAP ranking — they correlate most directly with AI market activity across all segments.
- **R&D intensity** (R&D as % of GDP, ICT BERD) shows stronger attribution for Hardware and Software segments where innovation intensity is highest.
- **GDP-level features** (GDP growth, total GDP) act as macro baseline anchors, providing regime context for the statistical baseline corrections.
- The ML correction term is *additive* on top of the statistical forecast — a large SHAP value indicates a segment/year where the statistical model's errors are systematically explained by a macro feature the time-series model cannot capture.

Assumptions & Sensitivity

The following is a practitioner summary of key modeling assumptions. Full derivations, sensitivity analyses, and fallback rules are documented in `docs/ASSUMPTIONS.md`.

- **Assumption:** Annual data from 2010 onward is sufficient for trend estimation (~15 observations per segment). **If wrong:** Shorter history means wider confidence intervals and higher overfitting risk; ARIMA parsimony constraints (max_p=2, max_q=2) and AICc correction partially compensate but do not eliminate small-N uncertainty.

- **Assumption:** A structural break occurred around 2022 (GenAI surge) that requires explicit modeling. **If wrong:** Ignoring the 2022 break inflates residuals by mixing pre-GenAI and post-GenAI regimes; forecast accuracy degrades materially, and Phase 3 ML would learn the parametric shift as a spurious pattern.

- **Assumption:** The four AI segments (hardware, infrastructure, software, adoption) can be modeled independently with post-hoc aggregation. **If wrong:** Cross-segment correlations exist (GPU demand drives cloud infrastructure simultaneously); independent modeling underestimates forecast uncertainty and misses spillover-driven accelerations.

- **Assumption:** The PCA first principal component of proxy indicators (R&D/GDP, ICT exports, patents, researchers per million, high-tech exports) is a valid AI market activity index. **If wrong:** If the first PC captures something other than AI intensity (e.g., general economic development), market size estimates inherit systematic bias in both level and trend.

- **Assumption:** TRBC-classified listed companies are representative of the broader AI market. **If wrong:** Private AI companies (e.g., OpenAI pre-IPO, Anthropic, Mistral) are excluded; the TRBC-based revenue series likely understates true market size, especially post-2022 when private AI fundraising exploded.

- **Assumption:** AICc (not AIC) is the appropriate model selection criterion for ARIMA orders. **If wrong:** Standard AIC systematically over-penalises at $N < 30$ — selecting higher-order models that overfit; the AICc correction (adding $2k(k+1)/(n-k-1)$) prevents this on short annual panels.

- **Assumption:** OLS is the appropriate GDP share regression baseline, upgraded to WLS or GLSAR only when diagnostics require it (Breusch-Pagan $p < 0.05$ for heteroscedasticity; Ljung-Box $p < 0.05$ for autocorrelation). **If wrong:** Using OLS with violated assumptions produces inefficient parameter estimates; the diagnostic-driven upgrade chain ensures the estimator matches the error structure.

- **Assumption:** All preprocessing (StandardScaler for PCA) must be fit on training data only inside each CV fold. **If wrong:** Fitting the scaler on the full dataset leaks future distributional information into training; reported CV metrics would be optimistically biased and would not generalise to out-of-sample periods.

Mathematical Appendix

AICc Small-Sample Correction

Standard AIC is modified by Akaike's small-sample correction (AICc) for annual panels where n < 30:

```
AIC = −2ℓ + 2k
AICc = AIC + 2k(k+1) / (n − k − 1)
```

where k is the number of estimated parameters and n is the number of observations. With n=15 and ARIMA(2,1,2) (k=6), the AICc penalty is 21 points larger than AIC — effectively ruling out higher-order models unless the fit improvement is substantial.

n	ARIMA(1,1,1) k=4	ARIMA(2,1,2) k=6
15	+4.00	+21.00
20	+2.67	+9.33
50	+0.83	+1.94

Inverse-RMSE Ensemble Weights

Ensemble weights blend the statistical and ML model predictions:

```
w_i = (1/RMSE_i + ε) / Σ_j (1/RMSE_j + ε), ε = 1×10⁻¹⁰
```

The epsilon guard assigns near-maximum weight to any model with near-zero RMSE (perfect in-sample fit), while preventing division-by-zero. Weights are recomputed per segment from the corresponding CV RMSE values.

Additive Ensemble Blend

The final ensemble forecast is constructed additively:

```
ensemble(t) = w_stat × stat_pred(t) + w_lgbm × lgbm_correction(t)
```

where `lgbm_correction` is the LightGBM model's predicted value for the statistical residual at time t, not a full independent forecast. This architecture preserves the statistical model's extrapolation properties while correcting for structured macro-driven errors.

Confidence Interval Construction

Confidence intervals are derived empirically from temporal CV residuals. For each hold-out fold, forecast errors e(t) are collected. The CI at level α is:

```
CI_α(t) = [forecast(t) − q_{(1+α)/2} × σ_e, forecast(t) + q_{(1+α)/2} ×
σ_e]
```

where σ_e is the cross-validation residual standard deviation and q is the corresponding normal quantile. Separate LightGBM models are trained for lower and upper CI bounds to allow asymmetric interval widths reflecting nonlinear macro-feature interactions.

Chow Structural Break Test

For a break at observation t*, the Chow F-statistic is:

```
F = [(SSR_full − SSR_pre − SSR_post) / k] / [(SSR_pre + SSR_post) / (n −
2k)]
```

where SSR_full, SSR_pre, SSR_post are sums of squared residuals from OLS on the full, pre-break, and post-break samples respectively, k=2 (constant + trend). Under H_0 (no break), F ~ F(k, n-2k). Rejection at $p < 0.05$ confirms a statistically significant break at t*.

PCA Composite Construction

For segment i, let X be the (n × p) matrix of p standardized proxy indicators over n years.

```
X̃_ij = (X_ij − X̄_j^{train}) / σ_j^{train} [standardize on training fold
only]
X̃ᵀX̃ = VΛVᵀ [eigendecomposition]
z_i = X̃_i · v₁ [first PC score = AI activity index]
```

Explained variance ratio $EVR_1 = \lambda_1 / \Sigma\lambda_j$ is logged per CV fold. $EVR_1 > 0.5$ is required for the PCA composite to carry sufficient information for meaningful market size estimation.

Disclaimer & Notes

> This report was generated from real-time API data and hybrid econometric–ML models. All market size estimates are model outputs based on proxy indicators — direct measurement of "AI market revenue" from statistical agencies is not available. Confidence intervals reflect model uncertainty only; additional uncertainty from data quality, post-2024 structural breaks, and unknown macro shocks is not captured in the CI bands.

Known Limitations

- **Private company exclusion:** Pre-IPO AI companies are not in the LSEG TRBC universe. Market size may be understated by 20–40% in software/adoption segments (PitchBook/CB Insights estimates).
- **Proxy measurement error:** The PCA composite uses ICT-sector proxies as AI activity indicators. If the first PC captures general economic development rather than AI-specific dynamics, all estimates carry a systematic level bias.
- **Geographic coverage:** 19 economies covering approximately 85% of global AI activity. Emerging AI ecosystems are not covered, introducing a 5–15% understatement growing to 10–20% by 2030.
- **Segment overlap:** GPU revenue may be double-counted across Hardware and Infrastructure. Per-segment estimates should be interpreted with the documented overlap ranges in `config/industries/ai.yaml`.
- **Forecast horizon:** Forecasts beyond 2027 carry materially higher uncertainty. 95% CI bands at 2030 span a wide range — this is the honest model output, not a limitation to minimize.

Data vintage: 2024 | **Generated:** 2026-03-23 | This document contains forward-looking estimates based on historical data and statistical models. Past patterns do not guarantee future results. All dollar figures are in USD billions, constant 2020 prices. Source code available at: github.com/[repo] | Reproducible with: `uv run python scripts/run_reports.py`